

SPATIAL INTELLIGENCE ANALYSIS ON TOWER TYPICAL RESIDENTIAL FLOOR PLAN

BROJECT BRIEF

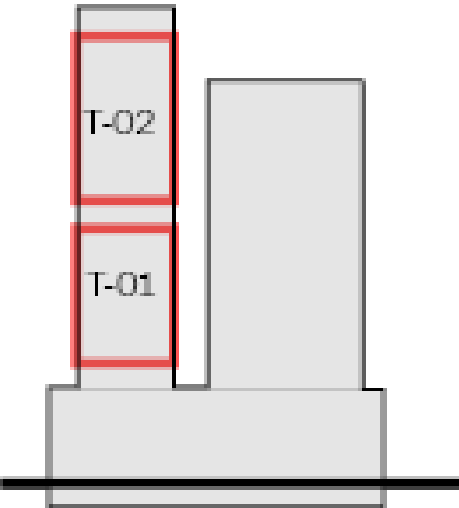
„ TYPICAL FLOOR PLAN OF RESIDENTIAL
TOWER CONSIST OF 10 TYPLOGY OF
APPARTMENT PER FLOOR WITH GROSS
FLOOR AREA 1350 SQM

Legend

- 3BDR+M
- 3BDR+M
- Balcony
- CIRCULATION
- CORRIDOR
- ELEVATORS
- FIREMAN LIFT
- SERVICES
- STAIRS

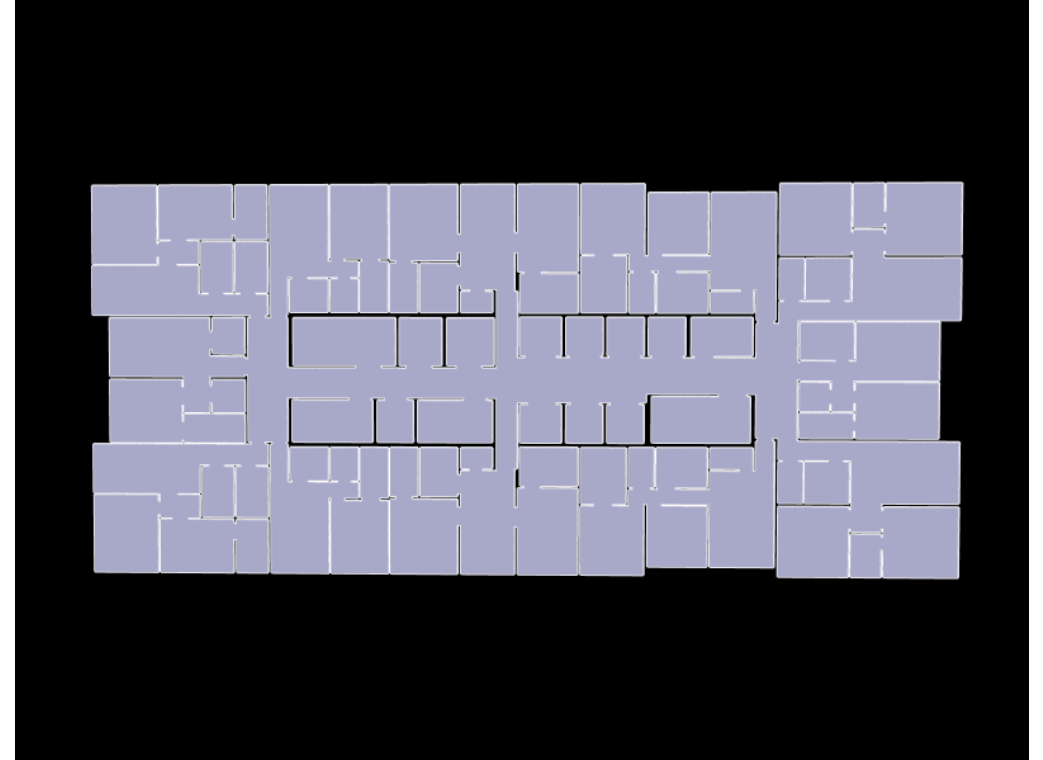
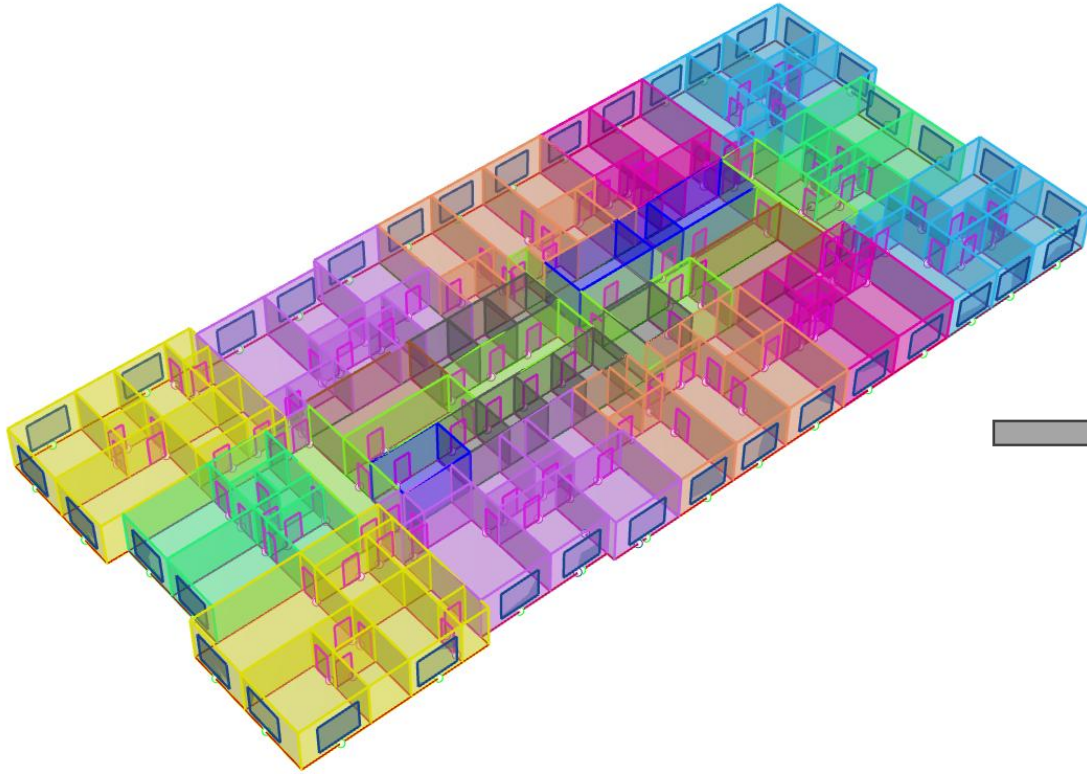


ARCHITECTURAL PLAN



SECTION

FROM OBJ TO .BREB



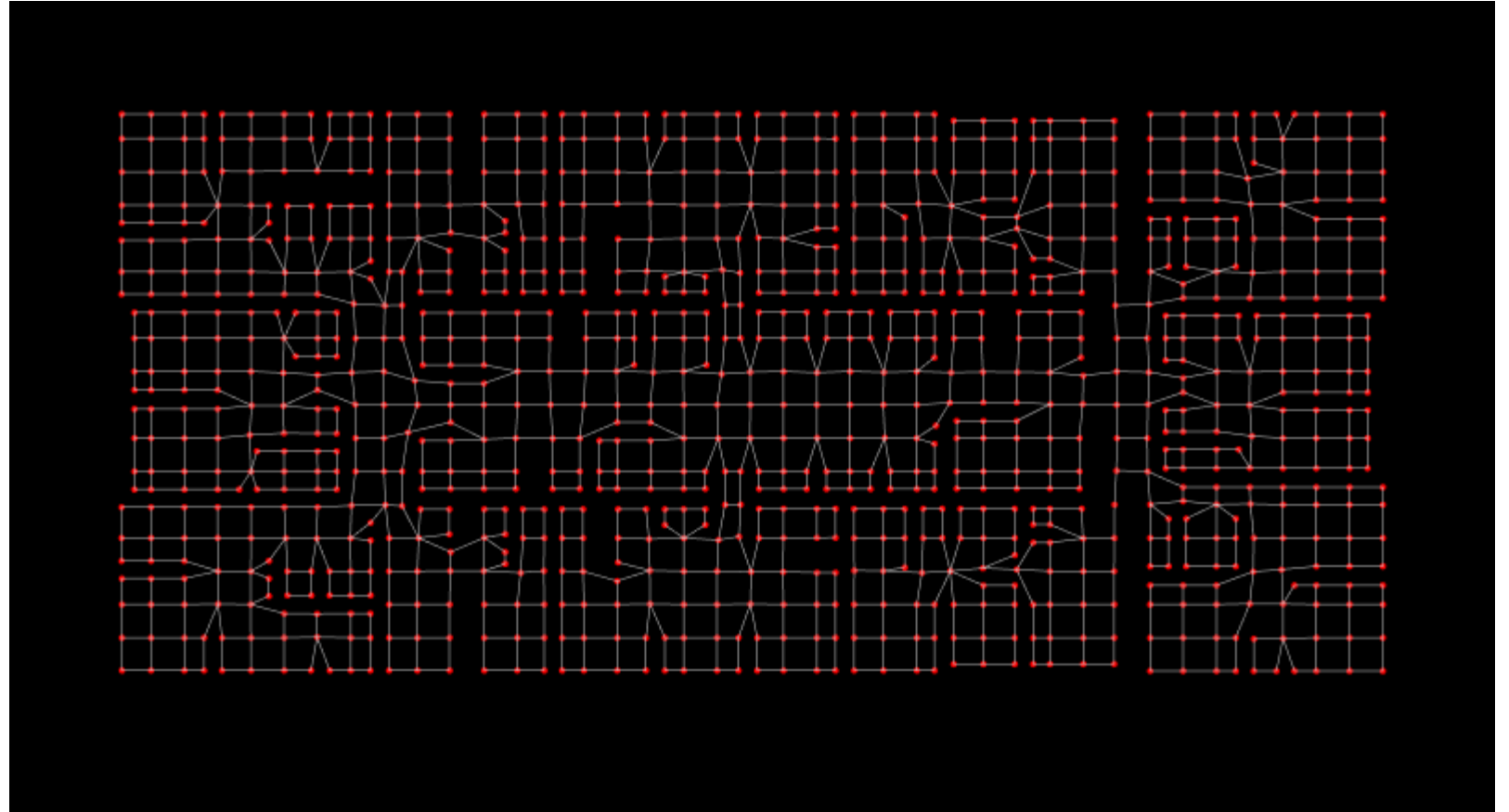
analysis graph vertices-spanning tree

```
grid_cell = 1.5  
uCount = int(width / grid_cell) + 2  
vCount = int(length / grid_cell) + 2
```

Density 1: 0.42857142857142855

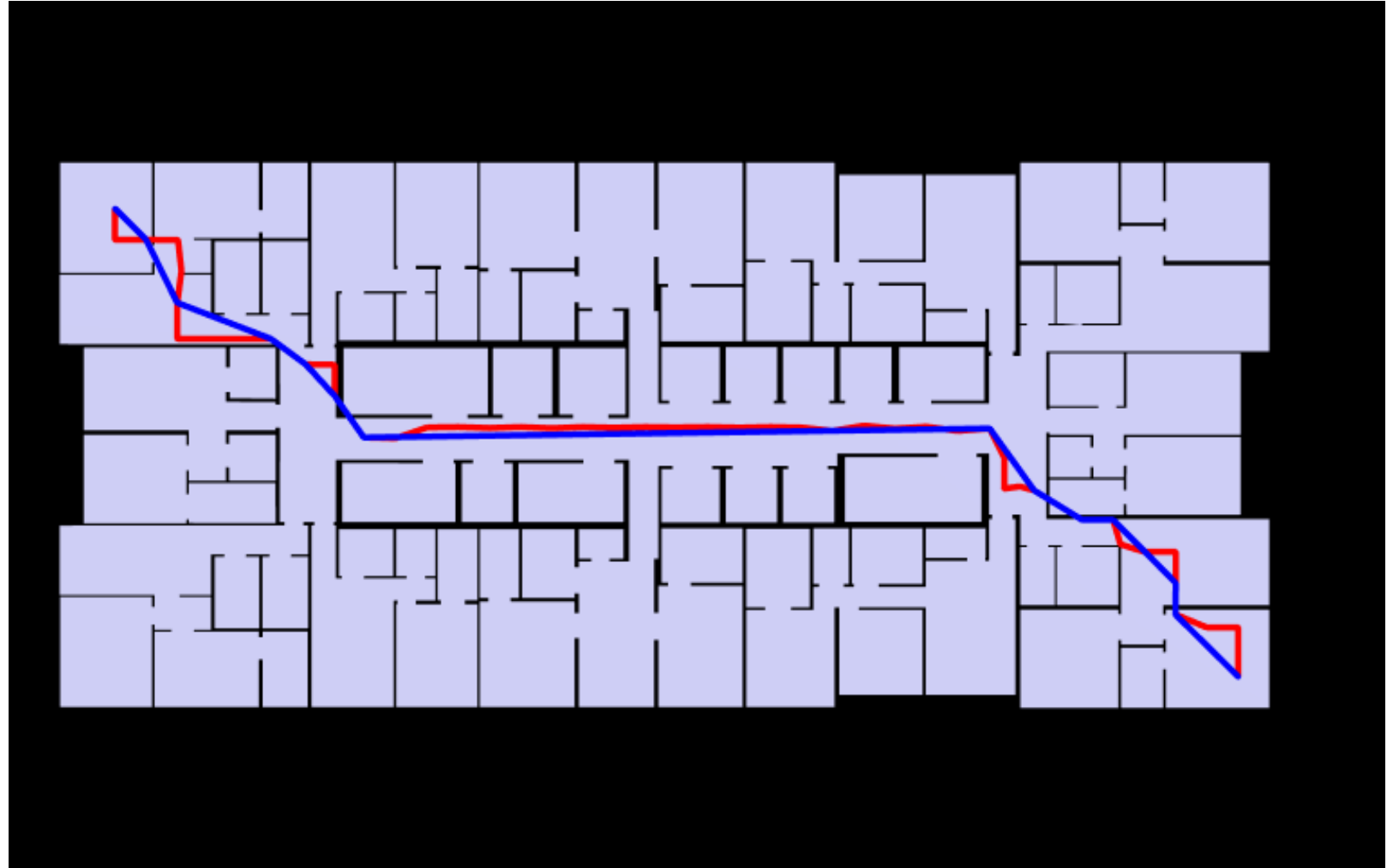
Density 2: 0.25

Density 3: 1.0



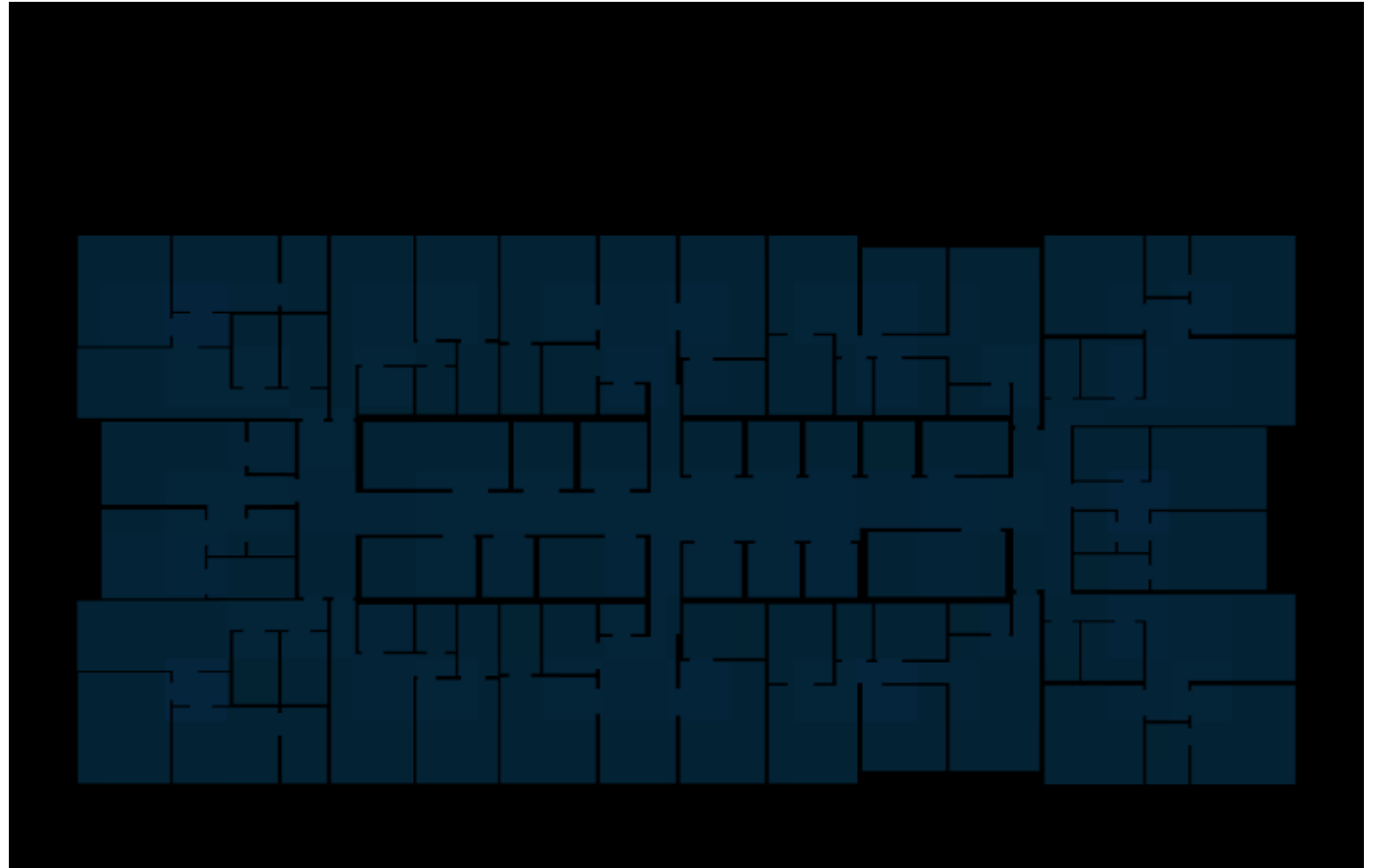
Shortest Path -Use navigation graph

Shortest Path Duration: 0.59 seconds
Straighten Wire Duration: 11.16 seconds
Original Shortest Path Length: 73.09
Straightened Shortened Path Length: 63.96



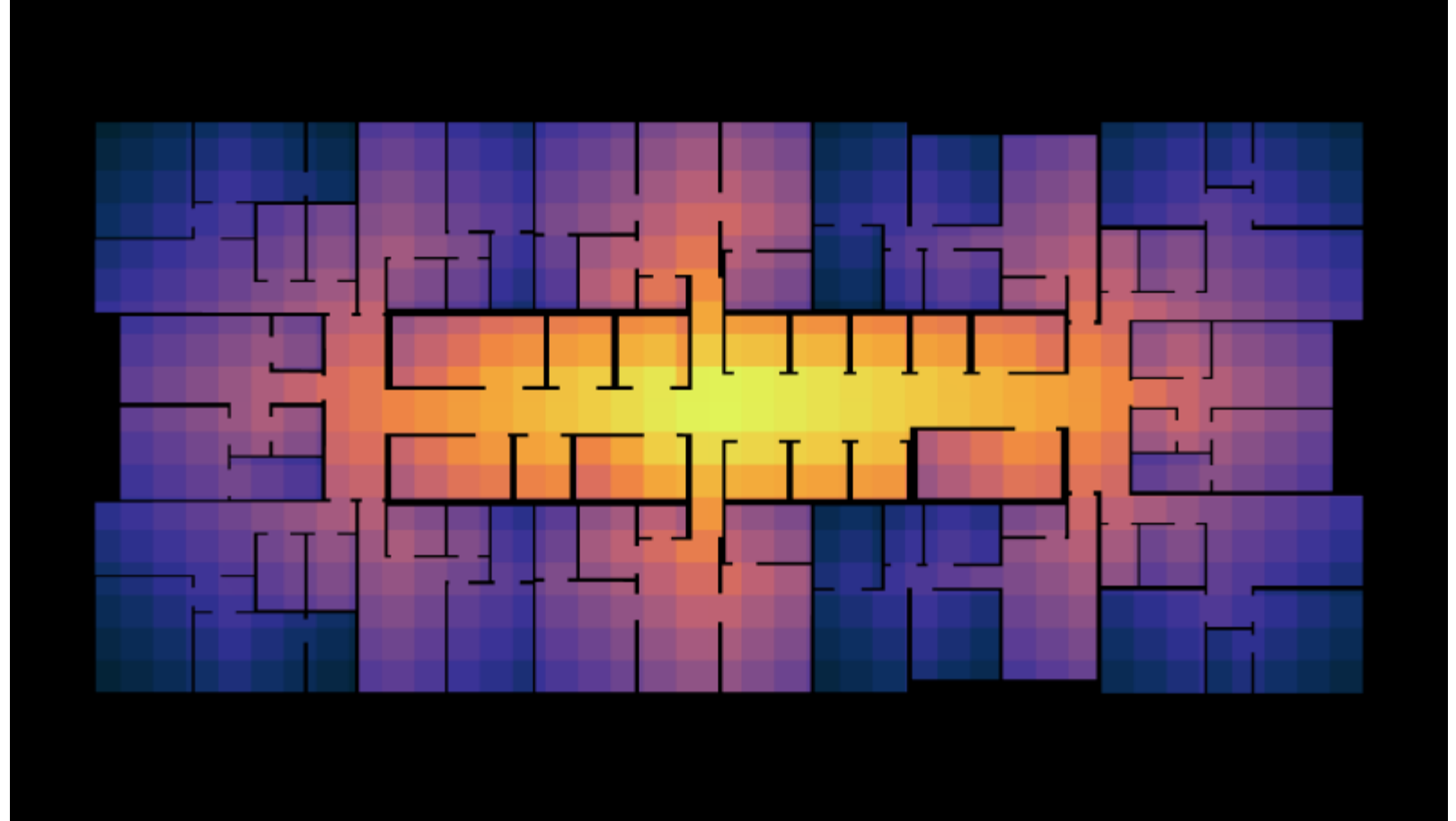
Degree Centrality/Connectivity

- * Degree centrality measures the number of direct connections (edges) a node has to other nodes in the graph.
- * In space syntax it is known as connectivity: Strong predictor of movement intensity at local scale



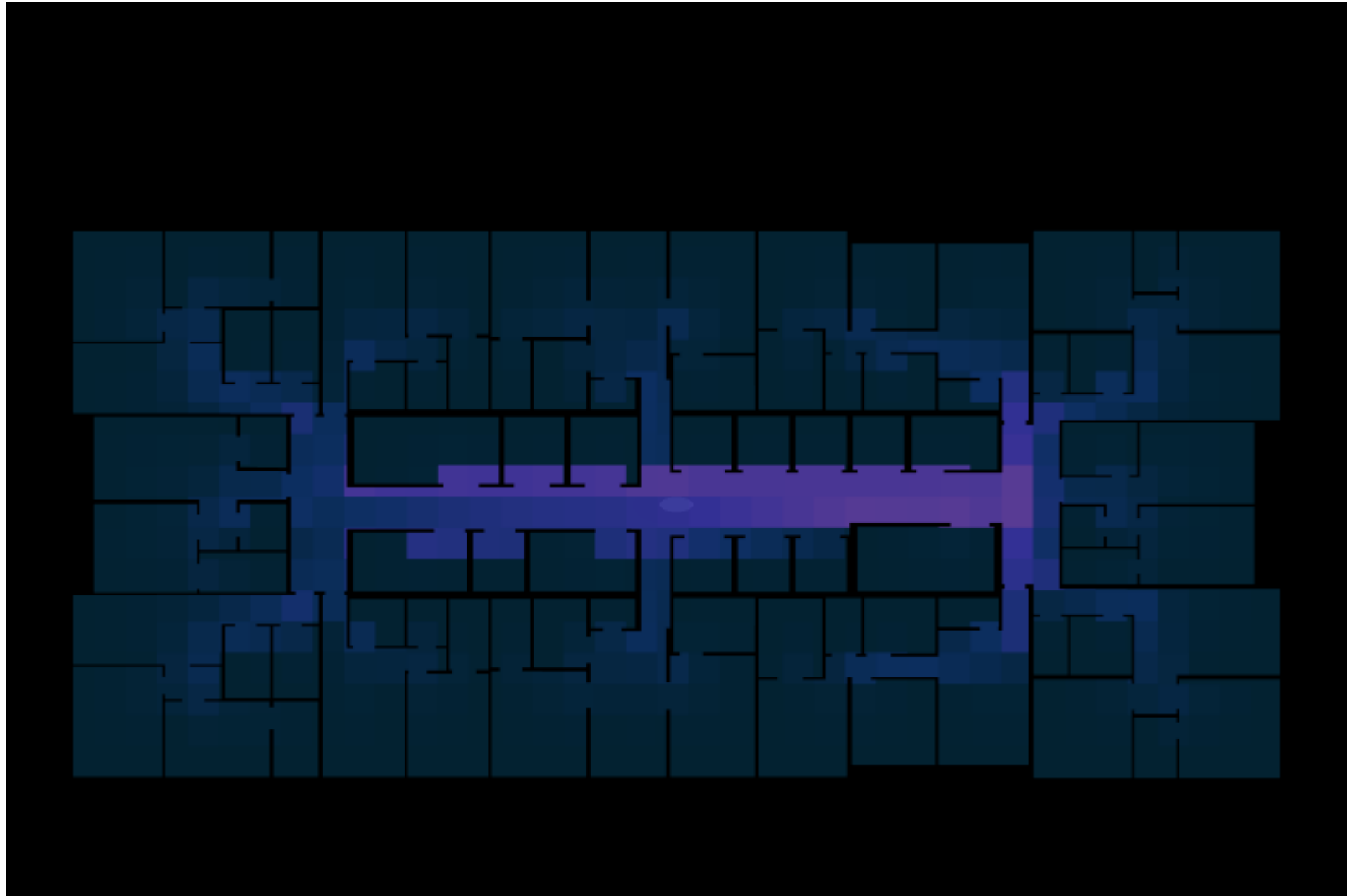
Closeness Centrality/Integration

- * Closeness centrality is a graph metric that quantifies how close a node is to all other nodes by taking the reciprocal of the sum of its shortest path distances to every other node in the network.
- * In space syntax, closeness centrality corresponds to global integration, measuring how spatially accessible or topologically shallow a space is within a configuration, thereby indicating its potential for movement flow and encounter density.



Betweenness Centrality/Choice

Betweenness centrality measures how often a node lies on the shortest paths between other nodes.



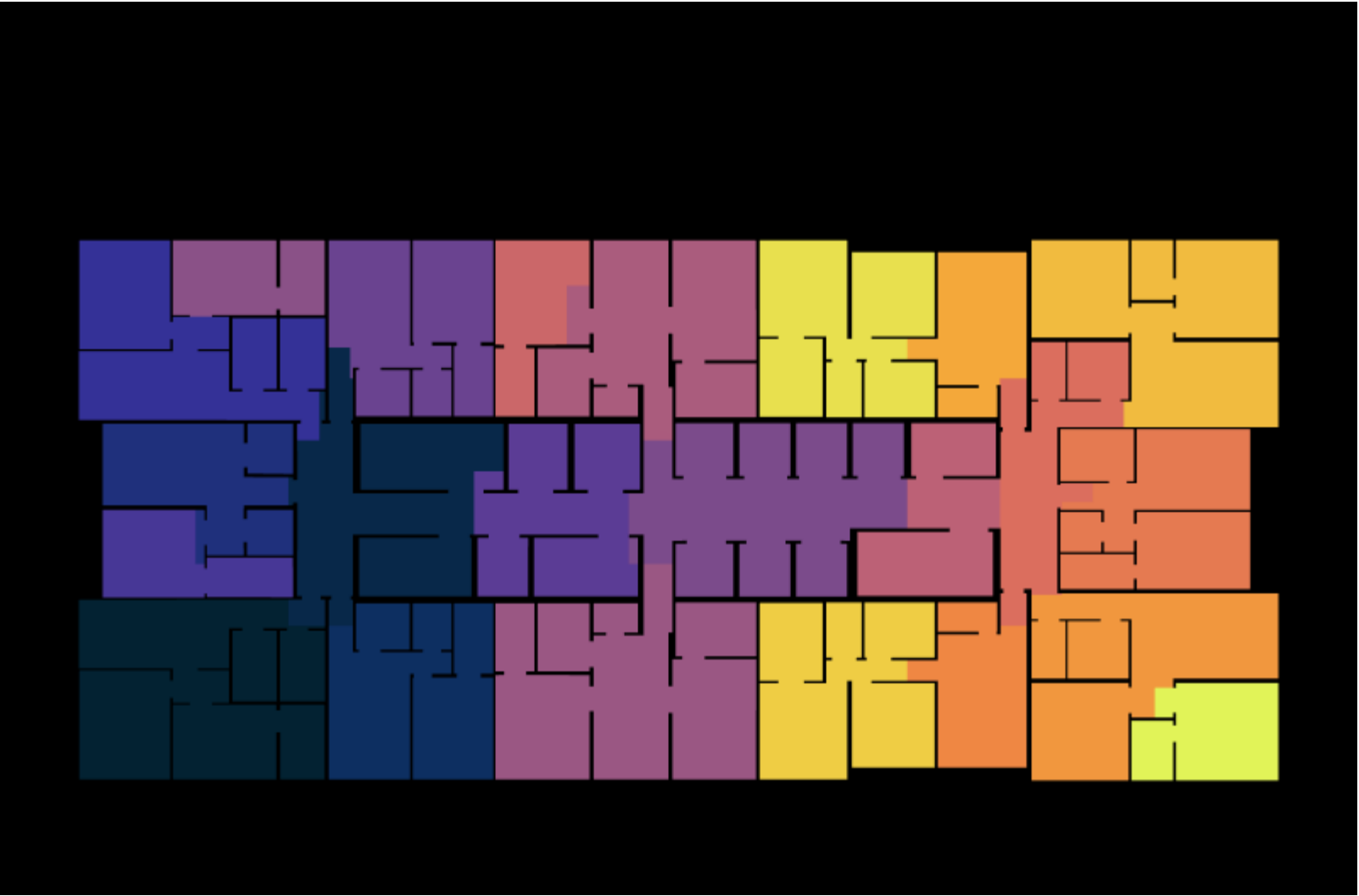
Community Detection

Communities: 24

Community		Area (m²)

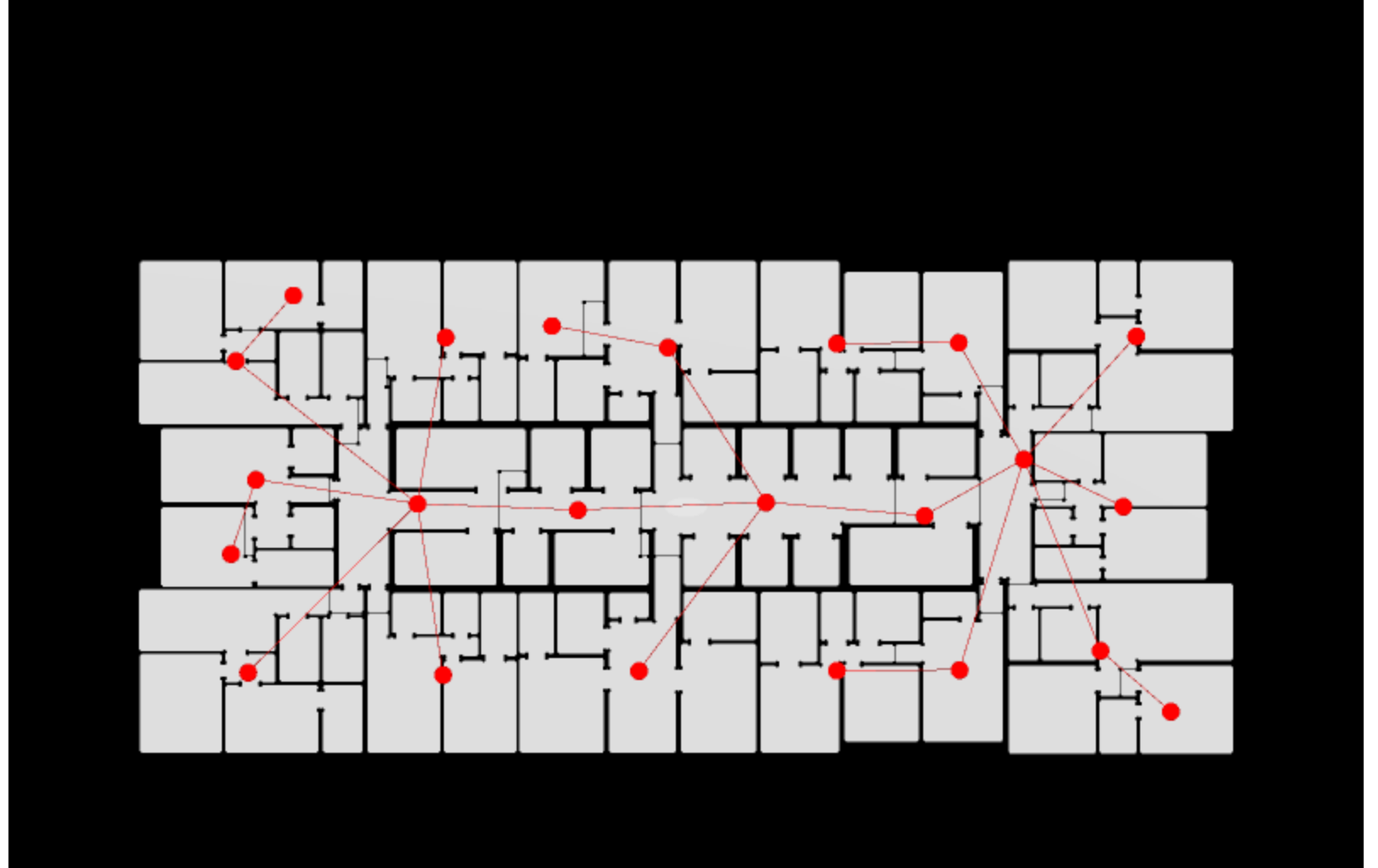
Community	11	105.87 m²
Community	0	97.94 m²
Community	13	82.84 m²
Community	12	75.24 m²
Community	8	69.97 m²
Community	5	67.75 m²
Community	17	67.17 m²
Community	20	65.93 m²
Community	22	65.93 m²
Community	2	64.57 m²
Community	7	62.12 m²
Community	21	56.72 m²
Community	23	56.61 m²
Community	15	55.87 m²
Community	1	52.67 m²
Community	14	52.12 m²
Community	4	49.44 m²
Community	3	46.44 m²
Community	18	41.35 m²
Community	16	35.88 m²
Community	19	33.02 m²
Community	6	28.04 m²
Community	9	26.31 m²
Community	10	25.07 m²

Total		1384.90 m²



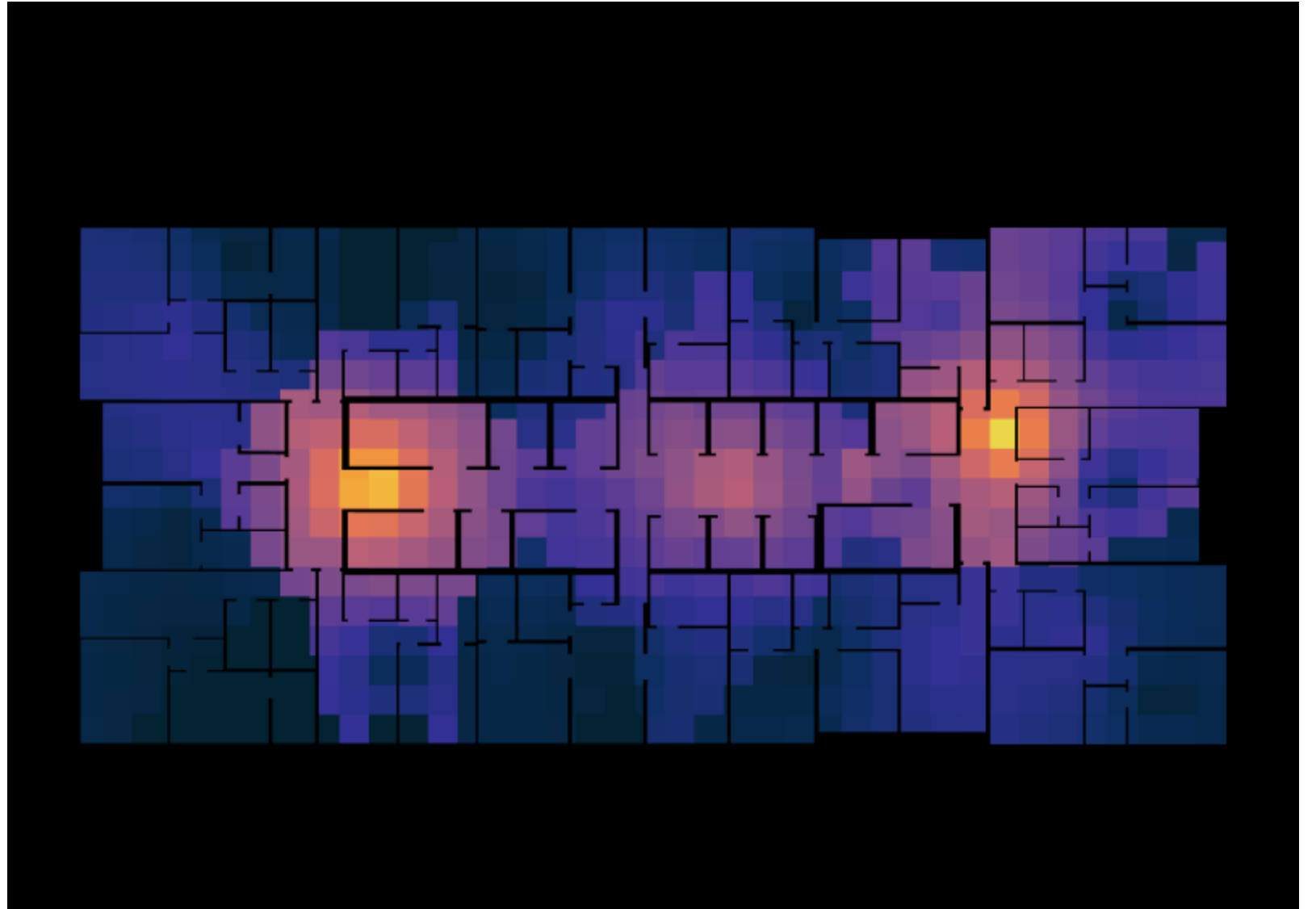
Degree centrality

derive the outer boundary of each face group
(perimeter) and make a face out of that perimeter



Degree centrality

Transfer the information from the graph
vertices to the faces of the original
shell



THANK YOU ,